

17. Hay que probar que $P(A \cap B) = P(A)P(B)$.

18. (a) En el avión de cuatro motores.

(b) Independencia de los motores.

19. 0.298

20. (a) No acierta. 0.7

(b) 0.09

(c) 0.42

(d) Independencia en los tiros.

(e) 0.09 y 0.35

21.

22. (a) $\{0, 1, 2\}$

(b)

$$F_X(x) = \begin{cases} 0 & \text{si } x < 0 \\ \frac{5}{12} & \text{si } 0 \leq x < 1 \\ \frac{11}{12} & \text{si } 1 \leq x < 2 \\ 1 & \text{si } x \geq 2 \end{cases}$$

23. (a)

$$p_X(k) = P(X = k) = \begin{cases} \frac{1}{36} & \text{si } k = 1, 12 \\ \frac{2}{36} & \text{si } k = 3, 11 \\ \frac{3}{36} & \text{si } k = 4, 10 \\ \frac{4}{36} & \text{si } k = 5, 9 \\ \frac{5}{36} & \text{si } k = 6, 8 \\ \frac{6}{36} & \text{si } k = 7 \\ 0 & \text{sino.} \end{cases}$$

(b) $E(X) = 7, V(X) = 5.83$

(c) 0.28

24. $E(X) = 39.28; E(Y) = 37$

25. (a) Ω tiene 8 elementos (falta describirlos)

(b) $\text{Rango}(X) = \{0, 1, 2, 3\}$ (falta asignar a cada elemento de Ω que valor le asigna X)

(c)

$$p_X(k) = \begin{cases} 0.125 & \text{si } k = 0, 3 \\ 0.375 & \text{si } k = 1, 2 \\ 0 & \text{sino.} \end{cases}$$

26.

$$p_X(k) = \begin{cases} 0.28 & \text{si } k = 0 \\ 0.27 & \text{si } k = 500 \\ 0.315 & \text{si } k = 1000 \\ 0.09 & \text{si } k = 1500 \\ 0.045 & \text{si } k = 2000 \\ 0 & \text{sino.} \end{cases}$$

27. (a) 0.5918

(b) No conviene jugar, la esperanza de X da -0.108

28. (a)

$$p_X(k) = \begin{cases} 0.5 & \text{si } k = 0 \\ 0.467 & \text{si } k = 1 \\ 0.03 & \text{si } k = 2 \\ 0 & \text{si no.} \end{cases}$$

$$F_X(t) = \begin{cases} 0 & \text{si } t < 0 \\ 0.5 & \text{si } 0 \leq t < 1 \\ 0.97 & \text{si } 1 \leq t < 2 \\ 1 & \text{si } t \geq 2 \end{cases}$$

(b) $E(X) = 0.533$, $V(X) = 0.3029$.

29. (a) (i) 0.103

(ii) 0.99886

(b) El valor esperado es 17.5 y la varianza es 5.25.

30. (a) 0.035

(b) 0.648

(c) 0.018

31. (a) 0.21

(b) 0.7

(c) 0.123

32. (a) 0.682

(b) 0.3958

(c) 0.0268

33. (a) 0.092 y 0.997

(b) $E(X) = 8$ y $\sqrt{V(X)} = 2.829$

34. (a) 0.143

(b) 0.053

(c)

$$p_X(k) = \begin{cases} 0.135 & \text{si } k = 0 \\ 0.2706 & \text{si } k = 1 \\ 0.2706 & \text{si } k = 2 \\ 0.18 & \text{si } k = 3 \\ 0.143 & \text{si } k = 4 \\ 0 & \text{si no.} \end{cases}$$

(d) 6

35.

$$p_X(k) = \begin{cases} 0.019 & \text{si } k = -100 \\ 0.073 & \text{si } k = -50 \\ 0.908 & \text{si } k = 0 \\ 0 & \text{si no} \end{cases}$$

y $E(X) = -5.55$

36. (a) $E(\hat{p}) = p$, $sd(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

(b) -

(c) -